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The promise and peril of using a large language model to obtain clinical information: ChatGPT performs strongly as a fertility counseling tool with limitations

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Objective: To compare the responses of the large language model-based "ChatGPT" to reputable sources when given fertility-related clinical prompts.
Design: The "Feb 13" version of ChatGPT by OpenAI was tested against established sources relating to patient-oriented clinical information: 17 "frequently asked questions (FAQs)" about infertility on the Centers for Disease Control (CDC) Website, 2 validated fertility knowledge surveys, the Cardiff Fertility Knowledge Scale and the Fertility and Infertility Treatment Knowledge Score, as well as the American Society for Reproductive Medicine committee opinion "optimizing natural fertility."
Setting: Academic medical center.
Patient(s): Online AI Chatbot.
Intervention(s): Frequently asked questions, survey questions and rephrased summary statements were entered as prompts in the chatbot over a 1-week period in February 2023.
Main Outcome Measure(s): For FAQs from CDC: words/response, sentiment analysis polarity and objectivity, total factual statements, rate of statements that were incorrect, referenced a source, or noted the value of consulting providers.
For fertility knowledge surveys: Percentile according to published population data.
For Committee Opinion: Whether response to conclusions rephrased as questions identified missing facts.
Result(s): When administered the CDC's 17 infertility FAQ's, ChatGPT produced responses of similar length (207.8 ChatGPT vs. 181.0 CDC words/response), factual content (8.65 factual statements/response vs. 10.41), sentiment polarity (mean 0.11 vs. 0.11 on a scale of -1 (negative) to 1 (positive)), and subjectivity (mean 0.42 vs. 0.35 on a scale of 0 (objective) to 1 (subjective)). In total, 9 (6.12%) of 147 ChatGPT factual statements were categorized as incorrect, and only 1 (0.68%) statement cited a reference. ChatGPT would have been at the 87th percentile of Bunting's 2013 international cohort for the Cardiff Fertility Knowledge Scale and at the 95th percentile on the basis of Kudesia's 2017 cohort for the Fertility and Infertility Treatment Knowledge Score. ChatGPT reproduced the missing facts for all 7 summary statements from "optimizing natural fertility."
Conclusion(s): A February 2023 version of "ChatGPT" demonstrates the ability of generative artificial intelligence to produce relevant, meaningful responses to fertility-related clinical queries comparable to established sources. Although performance may improve with medical domain-specific training, limitations such as the inability to reliably cite sources and the unpredictable possibility of fabricated information may limit its clinical use. (Fertil Steril® 2023;120:575–83. ©2023 by American Society for Reproductive Medicine.)
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Key Words: Artificial intelligence, natural language processing, fertility knowledge, counseling, online

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Since its inception, the internet has been recognized for its ability to provide quick and essential healthcare information for providers and patients (1). The world of reproductive medicine is no exception, with fertility-oriented content proliferating over the years. Investigators of a 2005 study highlighting the fact that a Google search for the word "infertility" yielded 1.8 million results (2) would be likely be amazed that the same search now returns nearly 1.1 billion (3). As much as this information—medically accurate or not (4)—has increased, the way patients and providers access the content of the web has remained fairly unchanged. Whether it is by accessing a search engine or browsing social media, infertility patients and their providers indicate a topic of interest and are shown a panoply of sources that they may consult. Interest is indicated in a variety of ways, whether it be by a search prompt or by lingering on a video to direct a social media app (5). However, to date, the only way to ask medically related questions and receive direct, reliable, and comprehensive answers has been to consult a healthcare professional. This threatens to change due to remarkable advances in the field of natural language processing (NLP), the branch of artificial intelligence (AI) concerned with giving computers the ability to learn from and use human language to communicate (6). The field has captured the attention of many with the introduction of the AI chatbot "ChatGPT" by OpenAI (7). ChatGPT—accessible through a publicly available website—may be thought of as a more capable version of the software that has been used on customer service websites for years that enables human users to have conversations with a computer interface. The unprecedented power of this new model, however, allows it to answer questions and follow language commands to produce novel text with exceptional range and detail (8).

ChatGPT's ability to perform language tasks such as answering questions, writing articles, or even telling jokes has in large part been fueled by the development of new deep learning algorithms (9, 10). One such algorithm, Generative Pretrained Transformer 3 (GPT-3) is notable for its immense training data set that included 57 billion words and 175 billion parameters from the "internet, books, and other sources" (11). Powered by this data and its algorithm, GPT-3 represented a leap in NLP performance in part due to its ability to perform well on so-called "few shot" or "zero-shot" tasks in which the model is given few or no examples by the user to guide its response (11). An example of a zero-shot task would be asking the model to write a paragraph in a scientific manner without also giving it a sample of what scientific writing looks like. ChatGPT was initially released in November 2022, using an updated version of the GPT-3 model ("GPT-3.5"). ChatGPT's chatbot format equipped the GPT-3.5 language model with an easy-to-use conversational interface, giving users the ability to generate fluent human-like text on demand (7). The public has taken notice, with ChatGPT being called the "fastest growing app of all time" (12) and acquiring over 100 million users in the 2 months since its release (13). ChatGPT marks a significant advance in the accessibility of AI to the average user because its only requirement for use is not programming experience but the ability to communicate in human language (14). Given the suddenly low barrier to utilization, it becomes easy to see how ChatGPT could be leveraged as a clinical tool for patients to access medical information.

The compelling prose of ChatGPT and similar large language models belies their danger. There are many limitations to using these models to access clinical information. As of February 2023, Open AI labels the chatbot "a research release (7)." It is limited to data no more recent than 2021 (15), does not currently cite sources (16), and has raised concerns about producing plagiarized (17) inaccurate (18) information. As ChatGPT does not have human judgment, it cannot "understand" when what it assesses as the most probable appropriate response is inappropriate or even incomprehensible. Researchers are studying the phenomenon that models like ChatGPT have been found to "hallucinate" (19) and generate text that is "nonsensical" or "unfaithful" to the provided source input (20). Concerningly, one industry observer suggests this may happen "15%–20%" of the time with ChatGPT, despite the model often sounding confident in its responses (21). As much as compelling answers can convince users that the model has insight and authority, ChatGPT, because of this writing, has neither (22). Nonetheless, the Pandora's Box of NLP has been opened and cannot be closed. Patients, enticed by ease of use and human-like language, will ask questions related to their care and receive quality answers often enough to keep coming back. That is why it is critical to characterize this model's performance as a clinical knowledge tool and understand its capability to mislead. We sought to use fertility as a domain to test ChatGPT.

MATERIALS AND METHODS

The study design is shown in Figure 1. The ChatGPT "Feb 13" version was tested across 3 domains during a 1-week period in February 2023 to test its competency in answering fertility-related clinical questions that a patient might consult the chatbot to answer. This study is exempt from institutional review board review because it did not include any interaction or intervention with human subjects or access to identifiable private information.

First Domain

The Centers for Disease Control and Prevention (CDC) publicly lists "frequently asked questions (FAQs)" about infertility on its website (23). Backed by the credibility of its name, the CDC provides a list of detailed answers to 17 questions such as "what is infertility?" or "how do doctors treat infertility?" The first domain consisted of entering these questions as prompts in ChatGPT during a single session. Answers were recorded, and characteristics were compared between CDC answers and the chatbot. It should be noted that ChatGPT can be prompted to produce answers with desired characteristics, such as indicating a length or style of response. Our aim here was to assess how ChatGPT would respond without such direction. Factual statements in ChatGPT's responses were counted and reviewed to determine whether they contained incorrect information and whether statements lacked a reference. Statements were considered incorrect when they made a statement that could be disproven, improperly categorized,

FIGURE 1

	Domain 1	Domain 2	Domain 3
Way of Testing ChatGPT	FAQs about fertility	Fertility Knowledge Surveys	ASRM Committee Opinion
What Testing Represents	Model's ability to handle what a patient is likely to ask	What the model "knows" relative to patient population	How well model reproduces clinical standard

Study design.

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inappropriately responded to the question, or had evidence of an algorithmic error, such as repeating the same statement twice. This interpretation of response data was reviewed by 2 physicians to ensure agreement with the classification. Sentiment analysis, which characterizes the nature of words in a given set of text, was performed with Python's TextBlob model, which produced a polarity score ranging from -1 (negative) to 0 (neutral) to 1 (positive) and a subjectivity score ranging from 0 (objective) to 1 (subjective) for CDC and ChatGPT responses (24). TextBlob was chosen because it is one of the more frequently used general models reported in healthcare literature (25–29) and is built on established Python libraries such as pattern and the natural language toolkit (30). For comparison, another popular language model—VADER (31)—was also applied to the analyzed text, and those results are included in the appendix. TextBlob works by breaking down a piece of text into words and phrases and then applying a machine learning algorithm trained on a large dataset of annotated text data (i.e., the model learns from its training data that words like "happy" are positive or that adverbs like "extremely" are associated with personal opinion and applies this to input text (32). Mean scores were compared with a two-tailed *t-test*. Proportions were compared using Fischer's exact test.

Second Domain

The second domain involved administering the validated fertility knowledge surveys, the Cardiff Fertility Knowledge Scale (CFKS) (33) and the Fertility and Infertility Treatment Knowledge Score (FIT-KS) (34) to see how ChatGPT would perform relative to patients. The CFKS is a questionnaire consisting of 13 items about basic fertility facts, misconceptions, and risk factors for impaired fertility, with an allowed response of true, false, or don't know. The FIT-KS contains 29 factual multiple-choice questions designed to assess knowledge of recent statistics (as of 2017) about natural fertility and treatment. Each question of each survey was entered as an individual prompt. The output was judged as correct if the response identified the true answer and incorrect if an alternative answer was indicated. Responses that did not clearly identify an answer choice were marked as inconclusive. Whether ChatGPT included an explanation for answer choices was noted. A score for each survey was calculated and compared to population data from Bunting et al.'s (33) international cohort of 10,045 men and women for the CFKS and Kudesia et al.'s (34) reported "phase 1" cohort of 127 US reproductive-age women for the FIT-KS. A z-test was used to generate a percentile score for ChatGPT's result for each survey. To ensure consistency, each survey was given 3 times during separate sessions by 2 physicians over 1-week in February 2023. Results were collected independently and compared.

Third Domain

For the final domain, we aimed to assess the chatbot's ability to reproduce the clinical standard in providing medical advice. Developed by the American Society for Reproductive Medicine (ASRM) Practice and Ethics Committee, committee opinions represent "expert consensus" and are reviewed for currency at least every 5 years (35). The committee opinion on "Optimizing Natural Fertility," released in 2022 (36) represents a useful subject to test as it provides practical advice that can be given to patients interested in fertility who are not

TABLE 1

ChatGPT vs. CDC responses to CDC FAQs about infertility.

Characteristic	CDC	ChatGPT	*P* value
Total responses	17	17	
Mean number of words per response	181.00	207.82	.58, NS
Mean polarity score (-1 = negative, +1 = positive)	0.11	0.11	.94, NS
Mean subjectivity score (0 = objective, 1 = subjective)	0.35	0.42	.12, NS
Total factual statements	177	147	
Mean factual statements per response	10.41	8.65	.46, NS
% ChatGPT factual statements incorrect	-	6.12%	
% ChatGPT factual statements with reference	-	0.68%	
% ChatGPT responses with more factual information than CDC	-	47.06%	
% Responses with reference to value of consulting a provider	41.18%	82.35%	.01

CDC = the Centers for Disease Control; FAQs = frequently asked questions; NS = not significant.

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TABLE 2

ChatGPT performance on the CFKS and FIT-KS fertility knowledge surveys.

Survey	Questions	% Correct	% Inconclusive	Percentile	% Answers with explanation given
CFKS	13	84.6%	0.0	86.9%[a]	100%
FIT-KS	29	75.9%	3.5%	95.1%[b]	72.41%

CFKS = Cardiff Fertility Knowledge Scale; FIT-KS = Fertility and Infertility Treatment Knowledge Score.
[a] According to published mean and standard deviation of Bunting et al. (33)
[b] According to published mean and standard deviation of "Phase 1" of Kudesia et al. (34)

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necessarily seeking treatment. All 7 summary statements from the document were rephrased as questions. For example, the statement "the chances of pregnancy and live birth for women significantly decrease after age 35 as the risks of aneuploidy and miscarriage increase" was rephrased as "after what age do the chances of pregnancy and live birth for women significantly decrease as the risks of aneuploidy and miscarriage increase?" Answers were marked consistent if ChatGPT correctly identified the sought-after information without providing conflicting text. As with the second domain, the questions were given 3 times on separate occasions by 2 physicians over 1-week in February 2023. Results were collected independently and compared.

RESULTS

ChatGPT performed strongly in all 3 domains. The first domain results are shown in Table 1. When administered the CDC's 17 FAQs about infertility, ChatGPT produced responses of similar mean length (207.8 words per response ChatGPT vs. 181.0 CDC, $P=.58$, not significant [NS]). In assessing the amount of factual content—which involved enumerating statements with verifiable information in both ChatGPT and CDC responses and averaging across the 17 responses considered—no significant differences were found (8.65 mean factual statements per response vs. 10.41, $P=.46$, NS). Sentiment analysis was effectively the same for both (mean sentiment polarity 0.11 vs. 0.11 respectively, $P=.94$, NS, mean objectivity 0.42 vs. 0.35, $P=.12$, NS). This would categorize both the typical responses from ChatGPT and CDC as "neutral" and of similar "subjectivity" in terms of the language used. Out of 147 factual statements by ChatGPT made in response to the FAQs, only 1 cited a source, ironically quoting the CDC in producing an incorrect "success rate for ART cycles" in patients <35 years of age. In total, 9 (6.12%) of the 147 (ChatGPT factual statements were categorized as incorrect. More often than the CDC, ChatGPT frequently referenced the value of consulting a provider to seek more information (82.4% of responses ChatGPT vs. 41.2% CDC, $P=.01$).

In the second domain (results shown in Table 2), ChatGPT achieved high scores compared to published population data on the validated fertility knowledge surveys. ChatGPT would have been at the 87th percentile of Bunting's 2013 international cohort of 10,045 men and women that completed the CFKS. For all 13 questions, ChatGPT provided context and justification for its answer choices. ChatGPT performed at the 95th percentile based on Kudesia's original 2017 "phase 1" cohort of 127 reproductive-age US women. For 8 (27.6%) of the 29 FIT-KS questions, ChatGPT provided only the answer without any further text. Among the 42 questions across the 2 surveys, ChatGPT only once produced an inconclusive answer, indicating neither the correct nor incorrect answer. Of note, results were consistent across all repeat administrations.

In the third domain, ChatGPT reproduced the missing facts for all 7 summary statements from the "Optimizing Natural Fertility" Practice Committee document when rephrased as questions with perfect accuracy. The statements and question phrasing are shown in Figure 2. In each of its 7 responses, ChatGPT highlighted the fact removed from the statement and did not provide any conflicting information. As with the second domain, results were consistent across all repeat administrations.

DISCUSSION

Progress in generative AI, which describes algorithms such as ChatGPT that can be used to create new content like text or images (37), has upended the general consensus on which jobs are most at risk of being replaced by computers in the near term (38). Seemingly overnight, those in creatively driven human professions such as writer (39), artist (40), or programmer (41) now face a world where many tasks can faithfully be completed by a computer. Clinicians, even in a procedural specialty such as reproductive endocrinology must take notice. It may be that traditionally human skills such as the ability to interpret and respond to personal questions are on par with areas such as embryology as lead applications for AI in reproductive medicine (42). ChatGPT's facility with language has already given it the ability to reach some of the standards that physicians are held to, such as passing the United States Medical Licensing Examination (43). Our study presents a snapshot of ChatGPT's impressive quality as a clinical tool at the time of this writing; however, it also highlights important limitations that should preclude its unmitigated adoption.

Answering Infertility FAQs from CDC

In answering the infertility "FAQs" from the CDC, ChatGPT produced answers of similar length, factual content, and sentiment. In almost half of the 17 questions administered, ChatGPT gave more facts than the CDC. Its error rate in

FIGURE 2

Committee Opinion	Question Posed to ChatGPT	All Answers Consistent?
The chances of pregnancy and live birth for women significantly decrease after age 35, as the risks of aneuploidy and miscarriage increase	After what age do the chances of pregnancy and live birth for women significantly decrease as the risks of aneuploidy and miscarriage increase?	Yes
The "fertile window" spans the 6-day interval ending on the day of ovulation. Frequent intercourse (every 1–2 days) during the fertile window yields the highest pregnancy rates, but results achieved with less frequent intercourse (2–3 times per week) are nearly equivalent. Couples should not be advised to limit the frequency of intercourse when trying to achieve pregnancy.	The "fertile window" spans how long and ends on what day of the menstrual cycle? What frequency of intercourse during the fertile window yields the highest pregnancy rates? Should couples be advised to limit the frequency of intercourse when trying to achieve pregnancy?	Yes
The use of fertility-awareness methods, such as ovulation detection kits and cervical mucus monitoring, has been shown to increase the probability of conceiving in a given menstrual cycle	Has the use of fertility-awareness methods, such as ovulation detection kits and cervical mucus monitoring, been shown to increase the probability of conceiving in a given menstrual cycle?	Yes
There is insufficient evidence that a specific diet or intake of particular macronutrients can improve natural fertility. Daily folic acid supplementation in women decreases the risk of neural tube defects in their children.	Is there insufficient evidence that a specific diet or intake of particular macronutrients can improve natural fertility? Does daily folic acid supplementation in women decrease the risk of neural tube defects in their children?	Yes
Specific coital positions and postcoital routines have no impact on fertility	Do specific coital positions and postcoital routines have any impact on fertility?	Yes
Alcohol abuse, recreational drugs, smoking, and high caffeine intake may all negatively impact fertility	How do Alcohol abuse, recreational drugs, smoking, and high caffeine intake impact fertility?	Yes
Higher exposure to certain endocrine-disrupting chemicals, such as polychlorinated biphenyls, and to air pollution may adversely impact fertility in women.	How does higher exposure to certain endocrine-disrupting chemicals, such as polychlorinated biphenyls, and to air pollution impact fertility in women?	Yes

Third domain: reproducing the summary statements from ASRM committee opinion "optimizing natural fertility." ASRM = American Society for Reproductive Medicine.

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producing facts was 6.1%, suggesting it can handle a range of common questions in an appropriately detailed and informative manner. Although it often warns users to consult a provider, ChatGPT does not indicate the level of uncertainty or flag when the model might be incorrect. It provides an illusion of reliability in its persuasive prose. Indiscriminate references to consulting a provider (seen in 82.4% of ChatGPT responses) do not help individuals tell when they can or cannot trust the model. Tech leaders have cautioned that "we may not notice" when ChatGPT-like bots go "horribly wrong," and our findings illustrate how that can occur (44). Concerningly, the one time it cited a source, ChatGPT produced an unfounded statement, raising questions about its ability to help promote misinformation.

Users cannot critically evaluate what ChatGPT says because, according to our analysis, it rarely, whenever cites its sources. Patients have cues to evaluate medical information from many existing internet sources that facilitate judgment of reliability (45), which are not present with large language model-based chatbots. For example, when entering the question "What is infertility?" into Google, a user gets a well-written answer clearly sourced from the Mayo Clinic, among other links (3). Patients can choose where to get their information and evaluate the source accordingly. In an environment where ChatGPT "directs" the search through the vast amount of information available on the internet, its inability to truly comprehend the meaning of a question asked of it may be laid bare. An example in our study was the listing of "zygote intrafallopian transfer" and "gamete intrafallopian transfer" in a response to a question about types of assisted reproductive technology. Although not incorrect, these procedures are becoming less common and unlikely to have relevance to the patient asking that question.

Taking Fertility knowledge surveys CFKS and the FIT-KS

The validated fertility knowledge surveys CFKS and FIT-KS have real-world data supporting their use and interpretation in humans. Early studies that have attempted to challenge ChatGPT by asking clinical questions are limited in their ability to meaningfully interpret the chatbot's response (46). Methods such as evaluating by expert interpretation of characteristics like "appropriateness" are inherently subjective. Studies proclaiming ChatGPT would pass standardized exams like the United States Medical Licensing Examination are abstracting this conclusion from its performance on publicly available practice questions (43). Even our first domain, FAQ responses, relied on some degree of subjectivity in assessing "incorrect" statements. This second domain of our analysis took advantage of ChatGPT's linguistic skills to give the model the same surveys the patients we are comparing the model's performance to receive.

Viewed from one angle, the high scores seen on the CFKS and the FIT-KS suggest the large language model's potential as a valuable clinical resource. In effect, at least this version of ChatGPT demonstrated it "knows" more than the average patient on fertility knowledge, at least according to published population data, highlighting its ability to inform the average user. Encouragingly, ChatGPT is a general model trained on nonspecific data. Theoretically, a model trained on reliable and up-to-date fertility-related information could perform even better than in our study. From another angle, this may be the most worrisome result possible. An often-wrong model would be easy for a patient to dismiss. Percentiles suggest ChatGPT falls in a good but not perfect area. The "uncanny valley" (47) describes the phenomenon that occurs when a machine or computer-generated image becomes so close to looking and behaving like a human that it becomes unsettling or disturbing to some people (48). ChatGPT's impressive performance has arguably placed it in the equivalent of this valley as a clinical tool, with its unreliability at unpredictable times precluding true utility. In order for ChatGPT to be clinically useful, it needs to be practically perfect.

It is possible, however, that different patient populations may interact with ChatGPT in different ways. For example, ChatGPT may have more utility with younger generations, who may be more adept at distinguishing between meaningful and potentially incorrect information from a virtual interface. More work needs to be done on how real-life users interpret and value ChatGPT responses.

Reproducing the Summary Statements from ASRM Committee Opinion "Optimizing Natural Fertility"

Perhaps one of the most impressive aspects of ChatGPT's performance in our assessment was its ability to reproduce the summary statements of an ASRM committee opinion. Although our method of rephrasing statements as questions can be considered somewhat leading, ChatGPT reliably identified the missing piece of information in all 7 prompts. The fact that a generalized model could reproduce a variety of concluding statements from a document written by content experts illustrates the reach of the model. ChatGPT is very much not a fertility-oriented instrument any more than it is one trained on finance, economics, arts, or countless other subjects. Beyond advising patients, this may point to the large language model's potential as a resource for clinicians. An ability to answer questions relating to high-level clinical guidance suggests we may not be far off from a language model informing a clinician-oriented resource such as UpToDate (49).

This study has several limitations. We only evaluated one particular iteration of a large language model, the February 13 version of ChatGPT. Similar models, such as the recently announced Google Bard (50) and AI-powered Microsoft Bing (51), are likely to be released in the near term and provide alternatives for patients. The nature and availability of these models to the public are subject to change on a day-by-day basis. ChatGPT itself keeps changing, and as of mid-February 2023, OpenAI has released 5 updates to the model. On March 14, 2023, OpenAI announced that ChatGPT would be able to access an updated version of GPT-3.5 known as "GPT-4," claiming that it is "more reliable, creative, and able to handle much more nuanced instructions," meaning that ChatGPT may already do better as a clinical tool than we observed (52). The vast data that powers the language model in this study is opaque to the public and cannot be evaluated. OpenAI provides vague information on where the data behind ChatGPT comes from. Further, even if we could more thoroughly investigate the source data, the nature of the transformer model behind GPT-3 is that we cannot see how it is weighting individual sources. In "predicting" prompt responses, ChatGPT may be taking data from unreliable references. As such, it is unclear to what extent ChatGPT simply got "lucky" in producing appropriate answers. It is entirely possible ChatGPT is sourcing data from the CDC itself in creating answers that we compared to the CDC. It is also possible that the next iteration of the model and/or others will do worse than observed in the application investigated here.

To characterize the volatility in model response with various updates, we repeated a subset of questions from Domain 1, the validated surveys from Domain 2, and the summary statement questions from Domain 3 over a month after our original investigation. This analysis used an updated ChatGPT "Mar. 23 version." The model produced similar results (Appendix Table) to the original findings across the 3 domains. Of note, the model achieved high but not perfect scores on the CFKS (76.9% correct, 71.9% percentile) and the FIT-KS (82.8% correct, 98.7% percentile). This supports the idea that, as of this writing, the model remains in a potentially useful yet problematic area for patients in providing clinical guidance.

CONCLUSIONS

In summary, ChatGPT is a powerful demonstration of what large language models are already capable of as a tool for seeking fertility-related information. It represents the manifestation of burgeoning technology that may change how we deliver certain aspects of care. In their ability to personalize responses, large language models show promise for AI by helping increase access to understanding and managing fertility (53). There have already been efforts to apply AI at the point where patients interact with their healthcare providers. For example, providers have explored using AI to help improve patient medication management (54). ChatGPT has the ability to dramatically accelerate these efforts. Furthermore, as online content often shapes patient conversations in the office (55), patients may soon come to their doctor with questions informed by ChatGPT responses. Providers should advise their patients to exercise caution when using such tools. Such models are subject to change and will constantly be vulnerable to corruption of their input. Increased utilization of tools like ChatGPT will make them more attractive to bad actors who may look to manipulate results for profit or other motives. Because we have seen with internet sources such as Twitter or Facebook (56), there will be an active effort to shape what users see. The possible accumulation of more "bad" and unreliable sources that feed the algorithm has the potential to detrimentally affect model performance. We have no ability to assess to what extent this is taking place because of the black box nature of current algorithms. Thus, it is crucial to continually reevaluate new and existing models. Clinicians may differ in their capability to optimally counsel patients, but at the end of the day, an individual can only reach so many. A language model has the capacity to reach millions.

REFERENCES

1. Pallen M. Introducing the Internet. BMJ 1995;311:1422–4.
2. Huang JYJ, Discepola F, Al-Fozan H, Tulandi T. Quality of fertility clinic websites. Fertil Steril 2005;83:538–44.
3. Google Search Results for "Infertility" and "Fertility". Available at: http://www.google.com. Accessed February 20, 2023.
4. Sun Y, Zhang Y, Gwizdka J, Trace CB. Consumer evaluation of the quality of online health information: systematic literature review of relevant criteria and indicators. J Med Internet Res 2019;21:e12522.
5. Kong W, Song S, Zhao YC, Zhu Q, Sha L. tiktok as a health information source: assessment of the quality of information in diabetes-related videos. J Med Internet Res 2021;23:e30409.
6. Russell SJ, Norvig P. Artificial intelligence: a modern approach. 4th ed. Pearson; 2021.
7. Open AI [Internet]. Available at: https://openai.com/. Accessed February 20, 2023.
8. Dale R. GPT-3: What's it good for? Nat Lang Eng 2021;27:113–8.
9. The Algorithm That Could Take Us Inside Shakespeare's Mind. New York Times. Available at: https://www.nytimes.com/2021/11/24/books/review/shakespeare-cohere-natural-language-processing.html. Accessed February 20, 2023.
10. ChatGPT can tell jokes, even write articles. But Only Humans Can Detect its Fluent Bullshit. Available at: https://www.theguardian.com/commentisfree/2022/dec/11/chatgpt-is-a-marvel-but-its-ability-to-lie-convincingly-is-its-greatest-danger-to-humankind. Accessed February 20, 2023.
11. Tingiris S. Exploring GPT-3: an unofficial first look at the general-purpose language processing API from OpenAI. Birmingham: Packt Publishing; 2021.
12. ChatGPT is the fastest growing app of all time. Available at: https://mashable.com/article/chatgpt-fastest-growing. Accessed February 20, 2023.
13. ChatGPT sets record for fastest-growing user base. Available at: https://www.reuters.com/technology/chatgpt-sets-record-fastest-growing-user-base-analyst-note-2023-02-01/#:~:text=Feb%201%20(Reuters)%20%2D%20ChatGPT,a%20UBS%20study%20on%20Wednesday. Accessed February 20, 2023.
14. van Dis EAM, Bollen J, Zuidema W, van Rooij R, Bockting CL. ChatGPT: five priorities for research. Nature 2023;614:224–6.
15. Duong D, Solomon BD. Analysis of large-language model versus human performance for genetics questions. medRxiv 2023:2023.01.27.23285115.
16. Biswas S. ChatGPT and the future of medical writing. Radiology 2023;307:e223312.
17. Stokel-Walker C. AI bot ChatGPT writes smart essays - should professors worry? Nature 2022.
18. Patel SB, Lam K, Liebrenz M. ChatGPT: friend or foe. Lancet Digit Health 2023;5:e102.
19. Bang Y, Cahyawijaya S, Lee N, Dai W, Su D, Wilie B, et al. A Multitask, Multilingual, Multimodal Evaluation of ChatGPT on Reasoning, Hallucination, and Interactivity. arXiv preprint arXiv:230204023 2023.
20. Ji Z, Lee N, Frieske R, Yu T, Su D, Xu Y, et al. Survey of hallucination in natural language generation. ACM Comput Surv 2023;55:1–38.
21. Got it AI creates truth checker for chat gpt hallucinations. Available at: https://venturebeat.com/ai/got-it-ai-creates-truth-checker-for-chatgpt-hallucinations/. Accessed February 20, 2023.
22. Shen Y, Heacock L, Elias J, Hentel KD, Reig B, Shih G, et al. ChatGPT and other large language models are double-edged swords. Radiology 2023;307:e230163.
23. Infertility FAQ's. Available at: https://www.cdc.gov/reproductivehealth/infertility/index.htm. Accessed February 20, 2023.
24. TextBlob: simplified language processing. Available at: https://pypi.org/project/textblob/0.9.0/. Accessed February 20, 2023.
25. Waudby-Smith IER, Tran N, Dubin JA, Lee J. Sentiment in nursing notes as an indicator of out-of-hospital mortality in intensive care patients. PLOS ONE 2018;13:e0198687.
26. Liu Z, Yang Y, Song H, Luo J. A prediction model with measured sentiment scores for the risk of in-hospital mortality in acute pancreatitis: a retrospective cohort study. Ann Transl Med 2022;10:676.
27. Kumar V, Bajpai R, Roy RB. Clinical notes mining for post discharge mortality prediction. IETE Tech Rev 2022;39:953–9.
28. Gao Q, Wang D, Sun P, Luan X, Wang W. Sentiment analysis based on the nursing notes on in-hospital 28-day mortality of sepsis patients utilizing the MIMIC-III database. Comput Math Methods Med 2021;2021:3440778.
29. Denecke K, Reichenpfader D. Sentiment analysis of clinical narratives: a scoping review. J Biomed Inform 2023;140:104336.
30. Loria S, Keen P, Honnibal M, Yankovsky R, Karesh D, Dempsey E, et al. Textblob: simplified text processing, Second. TextBlob Simpl. Text Process 2014;3:2014.
31. Hutto C, VADER GE. A parsimonious rule-based model for sentiment analysis of social media text. Proceedings of the international AAAI conference on web and social media 2014;8(1):216–25.
32. Microsoft. Documentation: text sentiment analysis. Available at: https://learn.microsoft.com/en-us/archive/msdn-magazine/2018/may/artificially-intelligent-text-sentiment-analysis. Accessed February 20, 2023.
33. Bunting L, Tsibulsky I, Boivin J. Fertility knowledge and beliefs about fertility treatment: findings from the International Fertility Decision-making Study. Hum Reprod 2013;28:385–97.
34. Kudesia R, Chernyak E, McAvey B. Low fertility awareness in United States reproductive-aged women and medical trainees: creation and validation of the Fertility & Infertility Treatment Knowledge Score (FIT-KS). Fertil Steril 2017;108:711–7.
35. ASRM Committee Opinion. Available at: https://www.asrm.org/news-and-publications/practice-committee-documents/. Accessed February 20, 2023.
36. Practice Committee of the American Society for Reproductive Medicine and the Practice Committee of the Society for Reproductive Endocrinology and Infertility. Optimizing natural fertility: a committee opinion. Fertil Steril 2022;117:53–63.
37. What is ChatGPT, DALL-E, and generative AI?. Available at: https://www.mckinsey.com/featured-insights/mckinsey-explainers/what-is-generative-ai. Accessed February 20, 2023.
38. Agrawal A, Gans J, Goldfarb A. Power and prediction: the disruptive economics of artificial intelligence. Boston: Harvard Business Review Press; 2022.
39. Bishop L. A computer wrote this paper: what ChatGPT means for education, research, and writing. SSRN Electron J 2023.
40. Frosio G. The artificial creatives: the rise of combinatorial creativity from dall-E to GPT-3. SSRN Electron J; 2023.
41. Kortemeyer G Could Artif-Intell Agent Pass Introductory Phys Course? 2023.
42. Zaninovic N, Rosenwaks Z. Artificial intelligence in human in vitro fertilization and embryology. Fertil Steril 2020;114:914–20.
43. Kung TH, Cheatham M, Medenilla A, Sillos C, De Leon L, Elepaño C, et al. Performance of ChatGPT on USMLE: potential for AI-assisted medical education using large language models. PLOS Digit Health 2023;2:e0000198.
44. Tech leaders warn that ChatGPT bots can make mistakes. Available at: https://fortune.com/2023/02/13/tech-leaders-google-search-chief-prabhakar-raghavan-apple-cofounder-steve-wozniak-chatgpt-chatbots-keep-making-errors/. Accessed February 20, 2023.
45. Breckons M, Jones R, Morris J, Richardson J. What do evaluation instruments tell us about the quality of complementary medicine information on the internet? J Med Internet Res 2008;10:e3.
46. Sarraju A, Bruemmer D, Van Iterson E, Cho L, Rodriguez F, Laffin L. Appropriateness of cardiovascular disease prevention recommendations obtained from a popular online chat-based artificial intelligence model. JAMA 2023;329:842–4.
47. Geller T. Overcoming the uncanny valley. IEEE Comput Graph Appl 2008;28:11–7.
48. ChatGPT & You: Language Generation's Uncanny Valley. Available at: https://medium.com/geekculture/chatgpt-you-language-generations-uncanny-valley-a91684325b0a. Accessed February 20, 2023.

49. UpToDate. Available at: https://www.wolterskluwer.com/en/solutions/uptodate. Accessed February 20, 2023.
50. Google AI updates: bard and new AI features in Search. Available at: https://blog.google/technology/ai/bard-google-ai-search-updates/. Accessed February 20, 2023.
51. Microsoft's bing chatbot offers some puzzling and inaccurate responses. Available at: https://www.nytimes.com/2023/02/15/technology/microsoft-bing-chatbot-problems.html. Accessed February 20, 2023.
52. Open AI. Releases GPT-4 AI that it claims is state of the art. Available at: https://techcrunch.com/2023/03/14/openai-releases-gpt-4-ai-that-it-claims-is-state-of-the-art/. Accessed February 20, 2023.
53. Swain J, VerMilyea MT, Meseguer M, Ezcurra D, Fertility AI Forum Group. AI in the treatment of fertility: key considerations. J Assist Reprod Genet 2020;37:2817–24.
54. Babel A, Taneja R, Mondello Malvestiti F, Monaco A, Donde S. Artificial intelligence solutions to increase medication adherence in patients with non-communicable diseases. Front Digit Health 2021;3:669869.
55. Tan SS-L, Goonawardene N. internet health information seeking and the patient-physician relationship: a systematic review. J Med Internet Res 2017;19:e9.
56. Schwartz LM, Woloshin S. Medical marketing in the United States, 1997-2016. JAMA 2019;321:80–96.

La promesa y el peligro de usar un modelo de lenguaje amplio para obtener información clínica: ChatGPT funciona fuertemente como una herramienta de asesoramiento de fertilidad con limitaciones

Objetivo: Comparar las respuestas del "ChatGPT" basado en modelos de lenguaje amplio con fuentes acreditadas cuando se les dan indicaciones clínicas relacionadas con la fertilidad.

Diseño: La versión del "13 de febrero" de ChatGPT de OpenAI se probó contra fuentes establecidas relacionadas con la información clínica orientada al paciente: 17 "preguntas frecuentes (FAQ)" sobre la infertilidad en el sitio web de los Centros para el Control de Enfermedades (CDC), 2 encuestas validadas de conocimiento de fertilidad, la Escala de Conocimiento de Fertilidad de Cardiff y la Puntuación de Conocimiento de Tratamiento de Fertilidad e Infertilidad, así como la opinión del comité de la Sociedad Americana de Medicina Reproductiva "optimizando la fertilidad natural".

Entorno: Centro médico académico.

Paciente(s): AI Chatbot en línea.

Intervención(es): Las preguntas frecuentes, las preguntas de la encuesta y las declaraciones resumidas reformuladas se ingresaron como indicaciones en el chatbot durante un período de 1 semana en febrero de 2023.

Principales medidas de resultado: Para preguntas frecuentes de los CDC: palabras/respuestas, polaridad y objetividad del análisis de sentimientos, declaraciones fácticas totales, tasa de declaraciones incorrectas, referencias a una fuente o el valor de los proveedores de consultoría.

Para encuestas de conocimiento de fertilidad: percentil según datos de población publicados.

Para la opinión del Comité: si la respuesta a las conclusiones reformulada como preguntas identificó hechos faltantes.

Resultado(s): Cuando se administraron las 17 preguntas frecuentes sobre infertilidad de los CDC, ChatGPT produjo respuestas de longitud similar (207.8 ChatGPT vs. 181.0 palabras/respuesta de los CDC), contenido fáctico (8.65 declaraciones / respuestas fácticas vs. 10.41), polaridad de sentimiento (media 0.11 vs. 0.11 en una escala de -1 (negativo) a 1 (positivo)) y subjetividad (media 0.42 vs. 0.35 en una escala de 0 (objetivo) a 1 (subjetivo)). En total, 9 (6.12%) de las 147 declaraciones fácticas de ChatGPT se clasificaron como incorrectas, y solo 1 (0.68%) declaración citó una referencia. ChatGPT habría estado en el percentil 87 de la cohorte internacional de Bunting en 2013 para la Escala de Conocimiento de Fertilidad de Cardiff y en el percentil 95 sobre la base de la cohorte 2017 de Kudesia para la Puntuación de Conocimiento de Tratamiento de Fertilidad e Infertilidad. ChatGPT reprodujo los hechos faltantes para las 7 declaraciones resumidas de "optimización de la fertilidad natural".

Conclusión(es): Una versión de febrero de 2023 de "ChatGPT" demuestra la capacidad de la inteligencia artificial generativa para producir respuestas relevantes y significativas a consultas clínicas relacionadas con la fertilidad comparables a las fuentes establecidas. Aunque el rendimiento puede mejorar con la capacitación específica del dominio médico, las limitaciones como la incapacidad de citar fuentes de manera confiable y la posibilidad impredecible de información fabricada pueden limitar su uso clínico.